



Emission enhancement in laser-induced breakdown spectroscopy from sample heating with a continuous-wave diode laser

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ABSTRACT

In this work an affordable, energy-efficient technique to enhance the sensitivity of the laser-induced breakdown spectroscopy technique is presented. This method is based in applying the beam of a continuous wave (CW) diode laser into the spot where the focused beam of a pulsed Nd:YAG, high-power laser induces the plasma. With the present set-up the CW beam heats the sample's surface to 300 °C. It was found that pre-heating the target in this way increased the line intensity of both neutrals and ions by a factor of up to 4.5. It was also found that the plasma excitation temperature and the electron density of the heated sample did not differ significantly from those at room temperature. However the amount of ablated material did change significantly when the sample was heated, with the crater in the latter case as much as 8.9 times larger than that obtained at ambient temperature. We conclude that the intensification in the pre-heated-sample case stems not from increased electronic plasma density or temperature but rather from a larger plume size resulting from an increased amount of ablated material when the target is heated.

1. Introduction

Laser-induced breakdown spectroscopy (LIBS) is a contemporary technique used for in situ composition analysis of solid, aerosol and liquid samples. The technique is based in spectroscopic analysis of the optical lines emitted by the plasma created by a high-power pulsed laser when it is focused on the target's surface [1,2]. This technique has several advantages like the fact that it does not require special pre-treatment of the sample, delivers rapid identification of species, etc. One of its disadvantages, though, is the fact that its sensitivity is of the order of parts-per-million (ppm).

For that reason, several enhancement techniques have been explored with the goal of increasing the sensitivity. Among the most frequently employed ones are: double-beam schemes (see, for example: [3–5]), electrical-discharge assistance [6–8], sample heating [9,10], among others [11]. The first two techniques can be configured so that the second pulse arrives in, either, orthogonal or collinear direction to the ablation laser [11–13]. This results in trace detection sensitivity improvements of an order of magnitude. In the case of pre-spark orthogonal and collinear configurations, the reduction in ambient density results in an enhancement of the removed mass. In addition, an increase

in plasma temperature and emission lifetime has been observed. As a consequence, the intensity of ionic lines increases more than the neutral ones. In the orthogonal reheating configuration the plasma temperature and the atomized mass also rise. Moreover, in both geometric configurations, an increase in the emitting volume has been reported.

In a report on laser ablation published by our group [10] it was found that by heating an aluminum sample in an oven the lines emitted intensify by an order of magnitude when the sample temperature increases from room temperature to 500 °C, under fixed laser energy. However in a separate experiment Tavassoli and Khalaji [14] found that for a copper sample heated with a resistor the increment of the line intensity is more modest, only 90% in going from 300 °C to 600 °C. They also found that at high target temperature both the ablated volume and the spectral emission increased.

Palanco et al. [15], reported the application of LIBS in steel casting technology. Their methodology involved the analysis, at a distance, of samples brought out of ovens with temperatures as high as 1200 °C. The analysis of their spectra showed that the high sample temperatures employed did not affect either the electron density nor the excitation temperature, both remaining constant throughout the temperature range employed. On their part, in an experiment where three different

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targets were heated up to 500 °C, Eschlböck-Fuchs et al. found that the sample temperature had no effect on the plasma's electronic temperature and density but it did affect the rate at which the plume expands [16].

The effect of heating a target with continuous-wave (CW) lasers on the ablation plasma properties has been investigated by a small number of researchers. For instance, Lednev et al. [17] used a high-power fiber laser, so powerful in fact that it could melt an iron sample, to investigate pre-heating effects. The authors found that the high temperatures did not have an effect on either the line intensity, electronic density and excitation temperature. It was only when the sample started to melt that these variables started to increase rapidly [17]. A somehow similar behavior was observed by Darbani et al., who investigated the effect of temperature on different alloys within the 215 K to 493 K interval and noticed that the electron density did not vary much within this interval but the excitation temperature increased noticeable as the target got hotter [18].

Shoursheini et al. found that the emission lines of the plasma induced by a Q-switched laser on a copper target were clearly affected by the concurrent application of a 20 W continuous-wave (CW) CO₂ laser beam [19]. Among the main effects observed by them are: some of the lines are intensified by a factor 9–19, the electron density basically remains unchanged with or without preheating and the electronic temperature is increased when pre-heating is applied. The goal of the present work is to further explore the effects that pre-heating with a CW laser has on the plasma induced by a Q-switched laser. In this work we will use a contemporary, affordable, energy-efficient laser diode to pre-heat the surface of a dielectric sample in order to investigate whether intensification can be achieved using a low-cost device.

2. Experimental set-up

The sample employed consisted of a 2-inch diameter clay pottery ceramic disc. The mineral content of the sample was: Si 32.7%; Al 8.8%; Fe 3.7%; Na 1.9%; K 1.4%; Mg 1.1%; Ti 0.6%, Ca 0.6%, Ba 0.01% plus other trace elements. The sample was locally heated by the beam of a continuous-wave laser diode (Laserland blue laser) that had a nominal wavelength of 450 nm and a square beam profile. One advantage of this diode is that it is stable and has a 15 kHz TTL switching ability. Its beam was slightly focused onto an area of approximately 3×3 mm at the surface of the sample by means of an $f = 10$ cm lens. The plasma was induced by focusing onto the previously heated spot the infrared (IR) beam of a high-power, pulsed, Q-switched Nd:YAG laser (Continuum Inlite III, 1064 nm, 7 ns pulse width) using a 10-cm focal length lens.

The light emitted by the plasma was collected by means of a collimator and sent through a fiber optics to an echelle spectrometer, as shown in Fig. 1. The field of view of this device spanned the entire length of the plasma plume, therefore the spectra obtained were spatially integrated. The spectrometer (Mecelle 5000) had a resolving power of 5000 and was coupled to an intensified-CCD camera. The spectral range covered by the spectrometer was 300–800 nm. In order to prevent the diode light from intruding into the measured spectra its operation was gated, in the following sequence: applied two minutes, then switched off 10 ms prior to an individual acquisition and switched back on again 10 ms after acquisition, ready for the next series. The timing of the pulsed laser firing, the spectrometer's gate and delay and the switching of the laser diode was controlled by means of a delay generator (BNC 575). Some acquisitions were time resolved, i. e. the acquisition was performed at different time delays after application of the laser pulse, others were time-integrated with a long gate (10 μ s) and a fixed delay of 500 ns.

The experiments were performed at the ambient pressure of our high-altitude lab (2200 m above sea level) that has an average air pressure of about 585 Torr. Two IR-laser pulses were applied to the sample before every test to clean the surface and then 20 more pulses applied on the same spot and their accumulated emission stored as a single spectrum. Between tests the sample was displaced ~ 1 mm so that

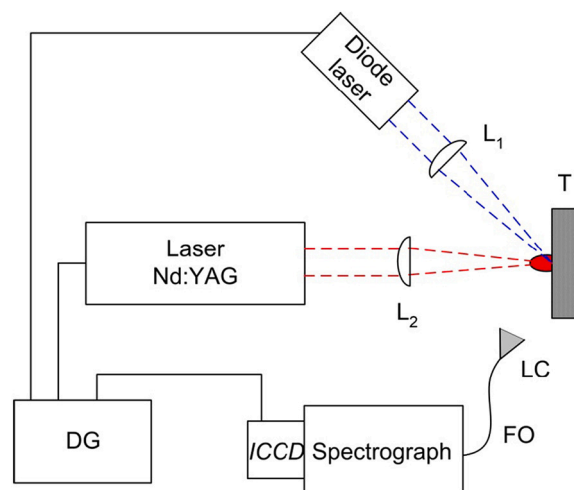


Fig. 1. Experimental set-up. Key: L₁, $f = 10$ cm plano-convex lens; L₂, $f = 10$ cm plano-convex lens; T, target; LC, light collector; DG, delay generator; FO, fiber optics.

each spectrum was taken at different, adjacent spots. In most of the tests the power of the CW diode was kept fixed at 5.5 W while the energy of the pulsed laser varied between 5 and 100 mJ. The diameter and depth of the craters was measured with a profilometer to estimate the amount of matter removed by ablation. It was found that, for example, for 15 mJ pulsed energy in the unheated case the crater diameter was 130 μ m which corresponds to a 1.6×10^{10} W/cm² irradiance.

In a limited series of experiments the expansion of the plasma plume and its associated shockwave was analyzed using the shadowgraphy technique. This was implemented as follows: the plasma was illuminated by a CW 532 nm laser beam (Dragon Laser, 532 M200) expanded by means of a 20 X beam expander and its shadow projected into a viewing screen. The resulting images were photographed with time resolution using a fast frame camera (Hamamatsu C2830) fitted with a 105 mm macro lens.

3. Experimental results

In the present work only a small section of the sample's surface was heated with the CW diode laser, the rest remained at room temperature. In order to assess the heating capacity of the device (without plasma) its beam was slightly focused onto an area of approximately 3×3 mm at the surface of the sample. Then the beam was applied steadily and as time elapsed the spot temperature was measured using an infrared gun thermometer (Fluke, 59Max+) that has a $\pm 1\%$ accuracy.

Fig. 2 shows the target temperature, without plasma, as a function of heating time for a fixed 5.5-W diode power. The graph shows that the temperature rises very quickly in the first ten seconds: from ambient to 216 °C. By time $t = 30$ s the temperature of the sample has already reached 83% of its maximum value. Thereafter, the temperature growth reaches a plateau and in fact the difference in temperature between an exposure time of two minutes and that of three minutes is only 2%. When the laser diode was turned off the sample temperature dropped quickly from 300 to 105 °C in half a minute.

Most of the experiments were made with a fixed diode power of 5.5 W applied during a period of two minutes which resulted in a spot temperature of 300 ± 8 °C at the sample's surface. The 3% estimated error in the temperature measurement is attributed misalignment of the pointer attached to the meter. We also monitored the air temperature in the vicinity of the sample using a K-type thermocouple. When the laser diode heated the sample to 300 °C (without Nd:YAG laser pulse) the air temperature was found to be 196 °C at a distance of 1 mm from the focus spot. The air temperature varied markedly in the upward direction,

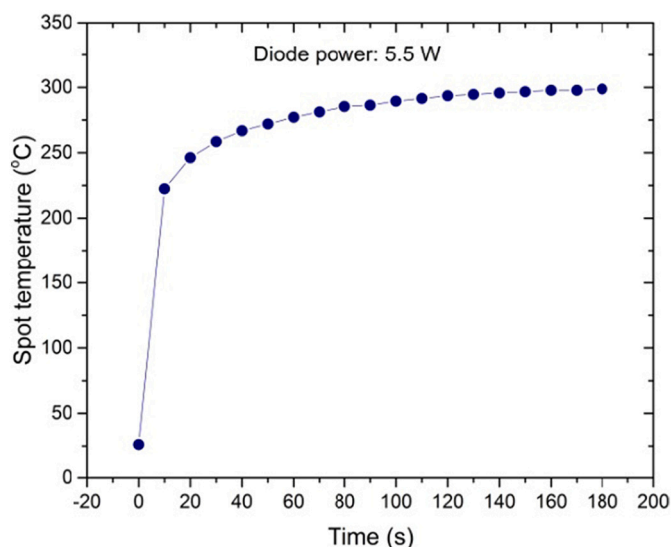


Fig. 2. Temperature rise at the sample surface produced by the laser diode. Fixed 5.5-W diode power.

parallel to the surface, due to convection heating; for instance, 6.5 mm above the spot the air temperature was found to be 83 °C warmer than the surrounding air.

Fig. 3 shows a portion of the spectrum produced by a plasma induced by the Q-switched laser operating at 50 mJ energy. The spectra were taken using a long gate, 10 μ s, which, for practical purposes, represents time-integrated acquisition. The laser diode beam was applied during 2 min prior to every individual acquisition, with fixed 5.5 W power that increased the sample's surface temperature to $\sim$ 300 °C, as shown in Fig. 2.

Shown in Fig. 3 are several lines of Ti II that clearly illustrate how the intensity of the lines is larger when the sample is heated by the CW laser in comparison with those at ambient temperature. For example, when heated, the intensity of the 336.12 nm line is two times larger than that at room-temperature. Similarly, Fig. 3 also shows that the intensity of the 339.45 nm, Ti II line is 2.1 times more intense than the corresponding one at room temperature. Neutral lines showed a similar behavior, for instance Fig. 3 shows that the height of the heated sample's 335.46 nm, Ti I line is 2.0 times larger than the room-temperature one.

In order to assess the effect that the power (P) of the diode has on the

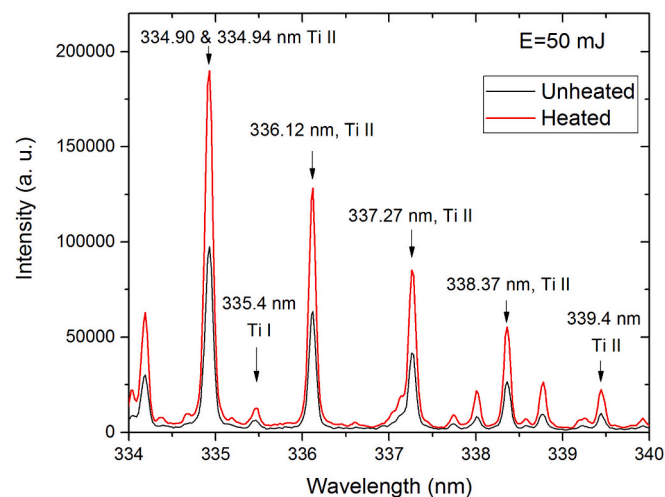


Fig. 3. Spectra obtained using 50 mJ Nd: YAG laser pulses, with and without target heating. Time-integrated acquisition.

intensity of the lines, a series of time-integrated spectra were acquired under variable CW laser power from 0 to 5.5 W keeping the heating time fixed at 2 min and also keeping the energy of the Q-switched laser constant. Here, the point $P = 0$ obviously corresponds to the unheated-sample case. Two lines of minority components, one neutral and the other a singly-ionized one were monitored.

Fig. 4 shows an example of the results obtained. It displays the intensity (the area under the curve, in this case) of two titanium lines as a function of the power of the diode laser, keeping constant the pulsed-laser energy at 15 mJ. In the figure it can be clearly seen that the lowest intensity of the Ti neutral line (499.1 nm) is obtained without pre-heating ($P = 0$ in the graph) and when the diode laser is switched on and its power increased the line intensity increases monotonically with P . The resulting intensification for the neutral line from no heating to maximum power is a factor of 1.9. The intensity of the singly ionized Ti line (336.1 nm) also increases with P so that from one extreme to the other it is intensified by a factor of 1.7. The error bars in Fig. 4 represent the standard deviation between six experimental runs carried out at different spots on the sample's surface.

It is important to assess the effect of increasing the laser energy not only on the line intensity but also on the noise level because if the signal increases and so does the noise, then the signal-to-noise ratio would not improve or could, even, deteriorate. In order to assess this aspect, the signal-to-noise ratio (SNR) was calculated using the area of the peak as the signal and the noise taken as the standard deviation of line-free zones in the spectrum. In a series of experiments the SNR intensification for main, minor and trace elements was investigated using time-integrated spectra. The experiments were made at fixed pulsed-laser energy and similar heating with the CW laser diode.

Fig. 5 shows the SNR values obtained for a selected minority-element transitions using a pulsed laser energy of 15 mJ. The bars in the figure are the ratio between the heated-sample intensity divided by the ambient temperature one. Two of the transitions that experienced larger amplification upon heating were: the 407.8 Sr I line that was intensified by a factor of 4.5 and the 643.9 nm Ca I line that increased by a factor of 3.7. The intensification of the previous lines is to be compared with the 2.6 factor intensification obtained for the 455.4 nm Ba II line. Note that, as mentioned in the previous section, calcium is a minority component in the sample as it makes 0.6% of the sample's contents so it is interesting that in spite of its low concentration, the present method successfully intensifies one of its lines (the 643.9 nm) by a factor of 3.75.

A series of time-integrated (10 μ s gate width) spectra were acquired to explore the effect that the Nd:YAG laser energy might have on the line

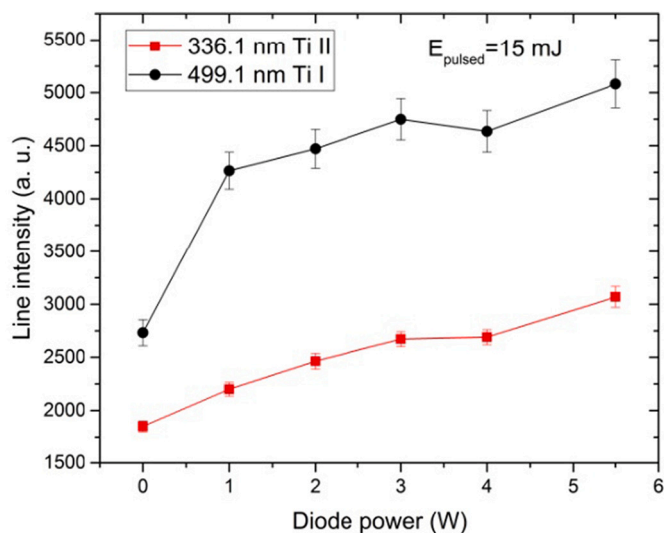


Fig. 4. Line intensity as a function of diode-laser power, fixed heating time. Nd: YAG laser energy: 15 mJ.

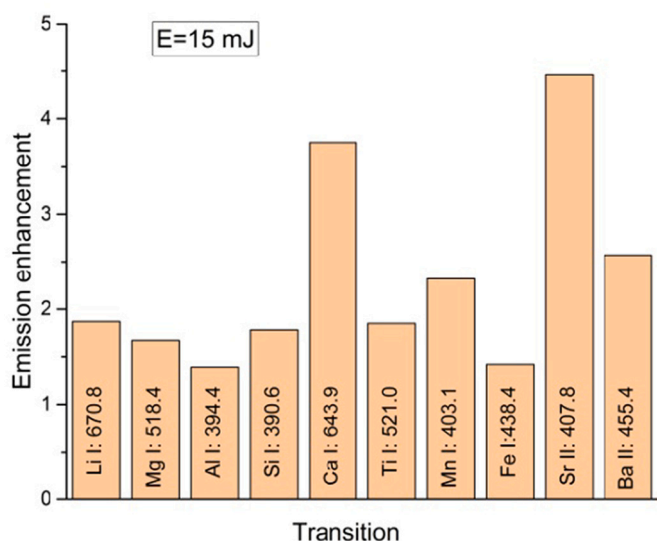


Fig. 5. Emission intensification for selected transitions due to pre-heating. Pulsed laser energy: 15 mJ. Transition wavelengths are in nanometers.

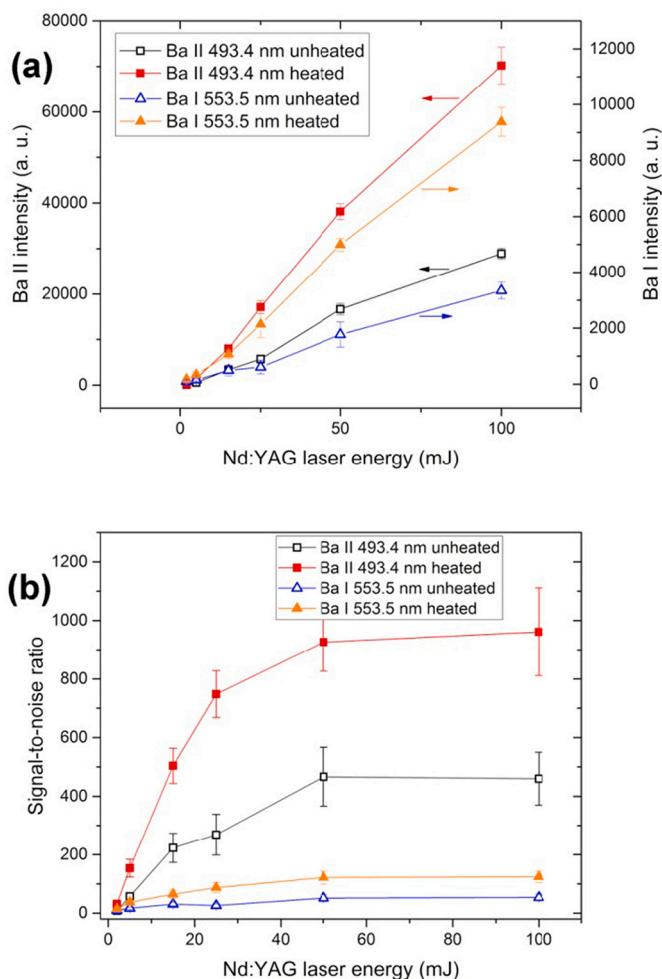


Fig. 6. a) Concurrent effect of pulsed-laser energy and heating on the intensity of the 553.5 nm, Ba I and 493.4 nm, Ba II lines, b) corresponding signal-to-noise ratio. Time-integrated acquisition.

intensity of trace elements under fixed diode heating (5.5 W, 2 min.). Fig. 6-(a) shows one example: shown there are the intensity of two barium lines, one neutral (553.55 nm) and the other an ionic one (493.4 nm) using different pulsed laser energies under identical heating (where applicable). The figure shows that the line intensity (the area of the peak in this instance) increases monotonically with increasing pulsed-laser energy and that, in fact, the intensity varies almost in a linear fashion with the energy.

Fig. 6-(a) shows that for a given pulsed-laser energy the intensity of any line clearly increases to twice or thrice its room-temperature value when the sample is pre-heated. In the case of the neutral line a maximum gain by a factor of 3.0 was obtained at a pulsed-laser energy of 25 mJ. In the case of the ionic line the intensity of the heated sample is a factor of 3.5 larger than the unheated case, also at 25 mJ pulsed-laser energy.

The signal-to-noise ratio (SNR) was calculated for the intensities in Fig. 6-(a) with the noise taken as the standard deviation of the line-free background. The results obtained are shown in Fig. 6-(b). For the two lines analyzed and the two conditions employed (namely, unheated and heated) the SNR increased very rapidly as the pulsed laser energy increased but the trend slowed at the higher energies of 50 and 100 mJ employed.

From the data in Fig. 6-(b) the enhancement caused by sample heating can be easily calculated from the division of the SNR of the heated case by that of the unheated one. For $E = 15$ mJ it turns out to be comparable for the ionic and the neutral lines, a factor of 2.3 and 2.1 respectively. The highest SNR intensification of the Ba lines in Fig. 6-(b) was a factor of 3.4 for the neutral line at $E = 25$ mJ.

In order to better understand the dynamics of the plasma with time resolution a series of shadow photography frames were taken at different delays after the instant of time when the Q-switched laser beam was applied ($t = 0$). Fig. 7 shows a sample of the results obtained when the sample was ablated using a 15 mJ pulsed-laser energy for the two

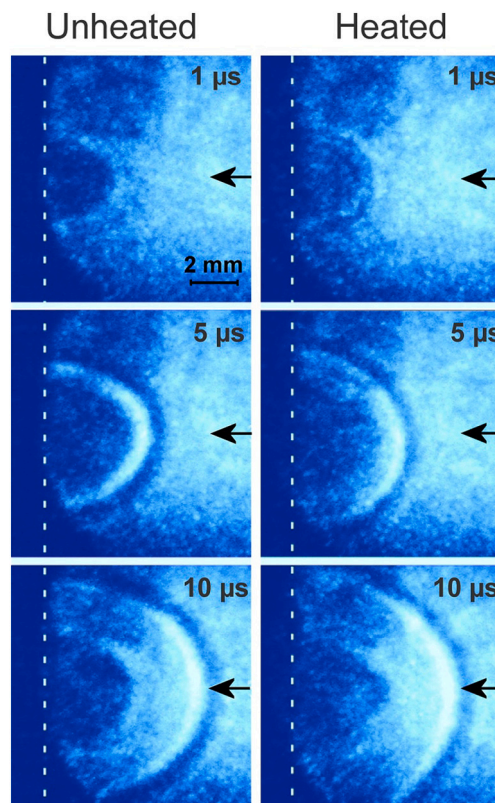


Fig. 7. Shadow photographs of the plasma with time resolution. The vertical dashed line indicates the position of the target's surface. The arrow indicates the direction of incidence of the incoming Nd:YAG beam (15 mJ). Gate: 20 ns.

cases: sample at room temperature and sample pre-heated to 300 °C. The column on the left-hand side of the graph shows how the plasma plume induced by the pulsed-laser without pre-heating and its shockwave evolve with time. The second column shows the expansion of both the plume and the shockwave in the preheated case as a function of time elapsed. In both cases the frames show concurrent expansion of the plume not only in the direction of the incoming IR beam but also in the transversal direction.

Fig. 7 shows that even though the shockwave expands at a similar rate in both cases, the plume behavior is different in each case: its size is larger in the pre-heated case, especially in the horizontal direction (length) rather than in the vertical one (width). For example at time $t = 10 \mu\text{s}$, the length of the unheated sample case the plume is 3.6 mm long whereas for the pre-heated one at the same delay the plume has a length of 4 mm. The plume expansion stagnates at $t = 5 \mu\text{s}$, this behavior is similar to that observed for a heated silicon sample in [16].

The use of the shadowgraphy images allowed the volume of the plume to be estimated with time-resolution. The volume was calculated under the assumption that the plume's envelope has an ellipsoidal shape. The results obtained for pulsed-laser energy of 15 mJ are shown in Fig. 8. As in the rest of this work, the sample was heated applying 5.5-W diode power for two minutes.

Fig. 8 shows that the size of the plume for the unheated sample increases in an almost linear way and reaches a maximum at $t = 5 \mu\text{s}$ and then decays a little bit. In the pre-heated case the size of the plasma grows at a faster rate than in the previous case and it also reaches maximum size at $t = 5 \mu\text{s}$. It is evident from the figure that the plume volume generated by the heated sample is consistently larger than the generated by the room-temperature sample; for example, at $t = 5 \mu\text{s}$ the former is 38% larger than the latter and at $t = 7 \mu\text{s}$ it is 50% larger.

The electron density of the plasma was calculated using the Stark broadening effect. The width of each line, and its corresponding error, was obtained from a numerical fit to a Lorentzian profile. Then the following, simplified, formula that relates the line width to tabulated data was used to obtain the electronic density [20]:

$$n_e = \frac{\Delta\lambda n_{ref}}{w_e} \quad (1)$$

where $\Delta\lambda$ is the line width (FWHM) minus the instrumental broadening, w_e is a tabulated width (FWHM) often called impact parameter, and n_{ref} is the density at which w_e was obtained. To obtain the density

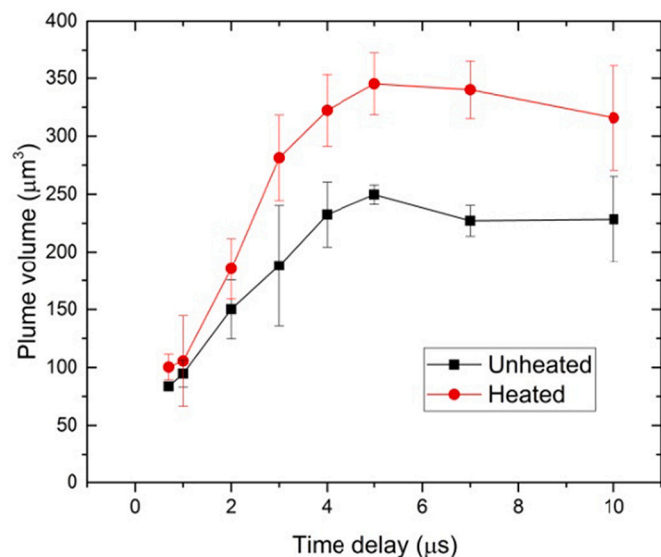


Fig. 8. Plume volume as a function of time for fixed (15 mJ) Q-switched laser energy for heated and unheated samples.

we used the 323.45, 324.2 and 337.28 nm Ti II lines and their parameters [21] as well as the 394.4 and 396.15 nm Al I ones [22].

The density was measured with time resolution through the application of the method described above to spectra obtained at different delay times, with reference to the pulsed laser application ($t = 0$), and with gates that varied from 200 ns for the shorter delays employed to 1 μs for the longer ones. Every spectrum represents the average of 20 individual acquisitions. Prior to any test the diode laser was applied for two minutes and then briefly switched off during an interval of 10 ms during which the pulsed laser was applied and the spectrometer triggered. This was done so in order to suppress the imprint of the diode laser on the measured spectra.

The electron density obtained for 15-mJ pulsed-laser energy with and without heating are shown in Fig. 9. The figure shows that for the unheated case the plasma density decreases rapidly in the first 2 μs , from a magnitude of $4.5 \times 10^{17} \text{ cm}^{-3}$ at $t = 0.5 \mu\text{s}$ to $6.6 \times 10^{16} \text{ cm}^{-3}$ at $t = 3 \mu\text{s}$ and subsequently remains more or less unchanged. In the case of the pre-heated sample, the initial density is 2/3 of the unheated one but at later times the densities in both cases are similar in magnitude and vary with time in a similar fashion. Measurements made at other pulsed-laser energies showed that there were not significant differences in density between the unheated and the heated cases, either with or without time resolution.

In 1968 W. Lochte-Holtgreven proposed that the temperature of a plasma can be obtained from the ratio of the intensity of two lines I_1 and I_2 emitted by atoms with different degrees of ionization through the following equation (re-written with a correction to a typo in [23]):

$$\ln\left(\frac{I_1 \lambda_1}{A_1 g_1}\right) - \ln\left(\frac{I_2 \lambda_2}{A_2 g_2}\right) = -\frac{E_1 - E_2 + \chi - \Delta\chi}{kT} + \ln\left[\frac{2(2\pi mk)^{3/2}}{n_e h^3} T^{3/2}\right] \quad (2)$$

where the index 2 designates the lower degree of ionization (neutral, for example) and 1 corresponds to the next higher one (singly-ionized in this example) $\lambda_{1,2}$ are the corresponding wavelengths, $A_{1,2}$ the transition probabilities, $g_{1,2}$ the degeneracies, $E_{1,2}$ each lines' upper levels, k the Boltzmann constant, m the mass of the electron, h Planck's constant, T the temperature, n_e the electron density, χ the ionization energy of the lower species and $\Delta\chi$ the correction to the ionization energy arising from Debye screening.

Eq. (2) is not amenable to graphical solution due to the fact that the temperature appears in two places on the right-hand side with different powers. To overcome this hurdle, Yalçın et al. [24] proposed a revised formulation that consists in displacing the logarithm as follows:

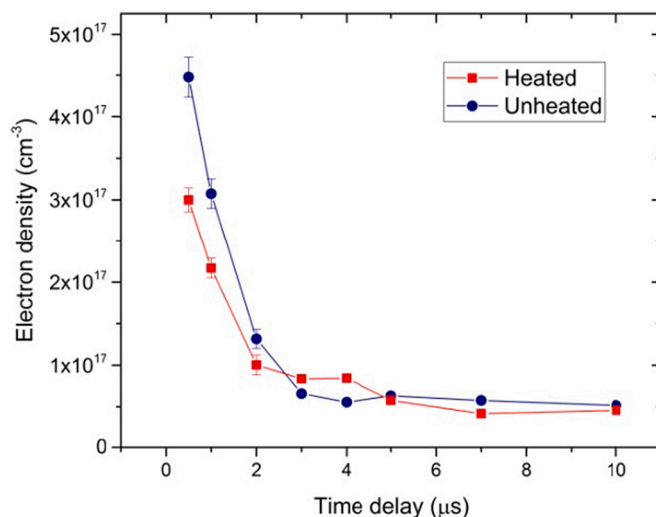


Fig. 9. Electron density as a function of time delay, Nd: YAG energy: 15 mJ.

$$\ln\left(\frac{I_1\lambda_1}{A_1g_1}\right)^* = \ln\left(\frac{I_1\lambda_1}{A_1g_1}\right) - \ln\left[\frac{2(2\pi mk)^{3/2}}{n_e h^3} T^{3/2}\right] \quad (3)$$

and similarly, the excitation energy:

$$E_1^* = E_1 + \chi - \Delta\chi \quad (4)$$

for species that are separated by one degree of ionization. Using this approach, the ionic lines can be plotted alongside the neutral ones in a plot where the y axis is the logarithm and the x axis is the E^* value. The resulting graph is known as Saha-Boltzmann plot and its slope is equal to $-1/kT$ and from it the excitation temperature of the plasma can be obtained through iterative calculations [25].

In order to obtain the excitation temperature with time resolution we applied this technique to spectra obtained at different delays relative to the application of the ablating laser pulse and using acquisition gates that varied from 200 ns to 1 μ s. The transition parameters required to apply Eq. (3) were taken from Ref. 26. The results obtained are shown in Fig. 10 for both the unheated and heated cases. In both cases the energy of the pulsed laser was kept constant at 15 mJ/pulse while the power of the diode laser was kept fixed at 5.5 W and the target was heated for two minutes. The figure shows that the temperature obtained with the target at room temperature is initially high at 11,200 K and afterwards decays exponentially as time elapses and flattens at the 7220 K level after $t = 5$ μ s in the unheated case. The error bars in the figure are the error in fitting the points in the Saha-Boltzmann plot to a straight line.

Also shown in Fig. 10 are the results obtained when the sample was preheated for two minutes prior to the application of the high-power plasma inducing laser pulse. In this case the temperature is, initially, slightly lower than that of the unheated case but as time elapses the two of them decrease with similar trend and have similar magnitude. It is evident from the plot that the preheating of the sample to 300 °C does not seem to have a significant effect on the excitation temperature: the differences in magnitude between the two curves in Fig. 10 are never more than 8%.

The size of the craters made by the pulsed laser pulse for both cases were measured using a profilometer. It was found that for all pulsed-laser energies employed both the diameter and the depth of the crater changed markedly when the sample was heated. Fig. 11 shows a comparison of the crater sizes obtained for a pulsed-laser energy of 15 mJ when the sample was at room temperature and when it was heated.

Fig. 11 shows that by heating the sample to 300 °C the diameter of the crater increased from 122 μ m at ambient temperature to 212 μ m

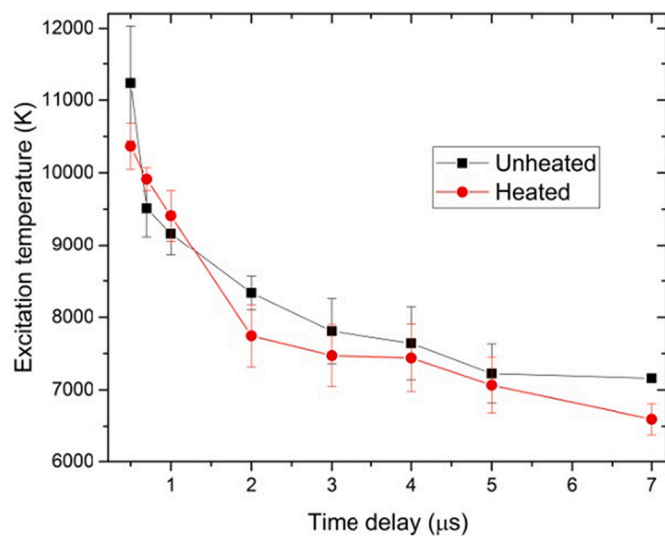


Fig. 10. Time-resolved excitation temperature, plasma induced with 15 mJ pulsed-laser energy.

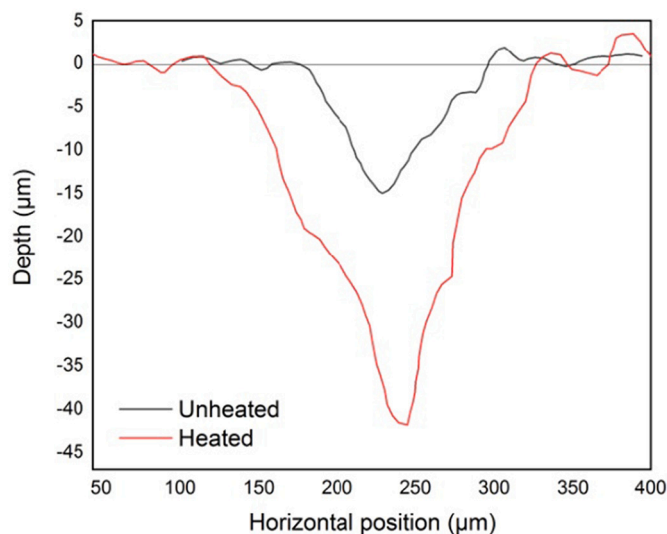


Fig. 11. Crater profile with and without pre-heating. 15 mJ pulsed-laser energy.

when heated while the corresponding changes in crater depth were 14.2 μ m unheated to 42 μ m heated. Assuming an approximately conical shape the volume of the crater was estimated to be 0.055 mm³ for the unheated sample and 0.49 mm³ for the pre-heated one, a factor of 8.9 increment for the latter case. Similar results were obtained at other pulsed laser energies, with increasing size as the IR laser energy increased.

4. Discussion

The diode laser proved to be an efficient way to heat at a fast rate (~ 100 °C/min.) the spot to be ablated on the sample surface. In fact, with applied 5.5 W power the target temperature rose, as shown in Fig. 2, in an almost step-wise fashion to ~ 300 °C in a matter of two minutes. Furthermore Fig. 4 shows that the maximum intensity is achieved at the maximum power available but also that satisfactory intensification can be achieved with 4 W diode power. The alternative methods for sample heating: oven baking or the use of a heating plate would have, easily, swallowed two orders of magnitude more electric power. Thanks to its energy efficiency, the diode laser is highly portable as it can be operated on batteries; this makes it suitable for field work as it can be used to improve the LIBS sensitivity for large samples such as buildings or rocks in remote sensing. It can also be used to analyze heritage samples such as archeological pieces for which bulk heating is unacceptable, but can tolerate spot heating that does not cause much damage.

LIBS users often apply other methods to increase the intensity and SNR of the lines emitted in order to improve the limit of detection. One of the more straightforward methods is to simply increase the energy of the ablating laser in such a way as to make the plasma hotter and therefore brighter. Fig. 6 shows that the method employed in the present work is an effective alternative to the use of higher pulsed-laser energies to increase the line strength. It can be readily seen from the data in Fig. 6-(b) that the SNR of both, the ionic and neutral Ba lines of the heated sample is clearly larger than the SNR of the unheated sample. Initially, in the interval 2 to 15 mJ, the intensities of the lines in Fig. 6(a) have a fast rate of growth with increasing laser energy, but after $E = 25$ mJ, in the interval 50–100 mJ, the rate slows markedly. This is so because at the latter energies the irradiance surpasses 10^{10} W/cm² and, as it is well-known [27], above this threshold the plasma is dense enough to become opaque and to reduce the amount of laser light that reaches the target so that the intensity of the lines no longer increases with laser energy at the same rate it did at lower irradiances. But, at the

same time, the noise continues to increase monotonically with increasing laser energy so that the net result is that the SNR flattens at the higher energies, as shown in Fig. 6(b). This effect is known as plasma shielding.

The enhancement depends on the transition being analyzed and the pulse energy. The variation of enhancement with pulse energy for ions and neutrals has a similar trend for both species. Fig. 4 shows that when the target is heated to maximum power the Ti I line enhances by a factor of 1.86 while the corresponding increment for the Ti ion is a factor of 1.66, a difference of 11% which is within the known variability of the LIBS technique. Also, the difference in enhancement between the barium singly-ionized and neutral lines in Fig. 6(a) is 13% on average.

Also evident from Fig. 6(b) is the fact that for the unheated target a five-fold increment in pulsed-laser energy from 5 to 25 mJ brings about a factor of 1.5 intensification of the neutral line's SNR whereas heating the sample with the CW laser increases it by a factor of 2.2 applying only 5 mJ of pulsed-laser energy. This result is replicated for other transitions of trace elements and is a clear demonstration that the method employed in this work is an effective means of improving both the SNR and the limit of detection.

Figs. 7 and 8 shows that the plasma plume expands very quickly in the first 5 μ s. Afterward, the growth stagnates and, in fact, decreases slightly. The envelope of the plume is consistently larger for the pre-heated sample, as much as 50% larger at $t = 7 \mu$ s. Most of this expansion is favored by a reduction in density of the air surrounding the heated spot. As mentioned above, the temperature 1 mm away from the sample's surface was measured to be 196 °C and this causes a reduction of relative air density to nearly half (49%) of the room temperature one.

Fig. 9 shows that at the early time of $t = 0.5 \mu$ s the electron density of the heated sample is 33% less than that of the unheated one, this is so because in the pre-heated case the plume is larger and contains more matter. However, because the temperature is not high enough, not all of the ejected material is ionized and, consequently, the electron density is reduced. As time elapses the difference in density between the two cases diminishes and in fact it reverses at $t = 3 \mu$ s but by then the plasma has cooled substantially, after $t = 5 \mu$ s the difference in the densities is insignificant.

Fig. 10 shows that there is no major difference between the time-resolved excitation temperature values for the ambient-temperature sample and the heated one as the difference between them never exceeds 8%. A similar trend was observed in the corresponding time-integrated measurements of T_e and n_e . In spite of the well-known variability of the LIBS technique, 10–15% differences between successive tests, this trend repeated itself in different runs. This is consistent with the results of other reports such as those in [10,16,17] where it is reported that the sample temperature has no meaningful effect on either T_e or n_e (although in [17] it is mentioned that heating has a small effect on n_e as the melting temperature is approached). On the other hand, Darbani et al. [18] found that the sample temperature did not affect T_e at all but had a clear effect on the density for some alloys.

The logic question arises: if the electronic temperature and density do not change upon application of local heating, then how do we explain the intensification of the lines observed? The answer is in Fig. 11, there the plots show a large difference between the room temperature case and the heated one, with a crater that is clearly wider and deeper for the heated sample. This means that there is more material removed in the heated target and therefore more ionized material and consequently more intensity as there are more emitters. However, the larger amount of material removed from the target's surface is distributed over a larger plume volume (see Figs. 7 and 8) so that the density does not change much with pre-heating.

5. Conclusions

Our investigation demonstrates that a CW diode laser represents an efficient way to heat the sample at the spot where the pulsed laser is

applied to generate the plasma. There is no need to heat the bulk of the sample (in an oven, for example) therefore making the present method a very efficient one, from an energy-efficiency viewpoint, to raise the sample temperature and consequently improve LIBS sensitivity.

Heating the sample with the laser diode turns out to be a more effective way to increment the line intensity than cranking up the pulsed laser energy. Preheating the target with the CW laser more than doubles not only the intensity but also the SNR for both the ionic and neutral species. When heating with the CW laser we obtained intensifications of line height of up to 4.5 times and a factor of 3.3 SNR intensification for minority elements. We found that the increment in line intensity can be attributed to more material injected into the plume and a larger plume volume that expands more easily into the reduced-density air in front of the heated spot.

As a way to intensify the emission, the CW diode laser has many advantages over other methods: it is energy efficient, it is portable (can be operated on small batteries) and therefore can be used on field work; it makes redundant the use of a large oven to heat the entire sample. Finally, with our technique we were able to intensify the plasma emission by a factor of 2 to 4 at a cost, at present time, of no more than \$100.

CRedit author statement

A. Robledo-Martinez: analysis of data, writing, investigation,
A. Garcia-Villarreal: investigation, analysis of data, conceptualization.
H. Sobral: investigation, analysis of data.
O. H. Magos: analysis of data.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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